Python - Day 1 - Assignment

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Batch: 2 to 4 PM
Course - Python full stack - Backend

1) Interpreter

* An Interpreter translates and executed code one statement at a time

(eg) python

Advantages

- easy to find errors
- easy to test and debug

Compiler

* A compiler translates the entire program with into machine code or Another target format

before execution

(eg) C

Advantages

- faster execution in many cases
- Detects errors during compilation

Difference

Interpreter	Compiler
- executes code through An Interpreter	- translates code before execution
- often easier to debug	- often faster during execution
- (eg) python	- (eg) C

2. OOP - object - oriented programming

* object-oriented programming is a programming approach that uses classes and objects to organize code.

(eg) A car is an object, and Car is its class

Main concepts of OOP

1. class : A Blueprint for creating objects
2. Object : An Instance of a class
3. Encapsulation : Combining data and methods in a class and controlling access to data.
4. Inheritance : one class can reuse the features of another class
5. polymorphism : the same method can behave differently for different objects
6. Abstraction : Hiding Implimentation details and showing essential features

Advantages of OOP

- code reusability
- Easy maintanence
- better code organisation
- Easy to manage large programs

(eg)

```
class Car:
    def start(self):
        print("Car is starting").

Car1 = Car()
Car1.start()
```

O/P

Car is starting

Example of OOP Languages

- python
- Java
- C++